Rachad El Moutaouaffiq

(438) 878-0603 | rachadelmtq@gmail.com | rachad.dev | LinkedIn | Canadian Citizen (TN Eligible)

Education

The University of British Columbia Sep. 2022 – Aug. 2027
Bachelor of Applied Science - Computer Engineering | *Dean's Honors List 2023-24-25* *Vancouver, BC*

Technical Skills

Software & Embedded: C/C++, Rust, Python, ARM Assembly, FreeRTOS, Linux, ROS 2, OpenCV, MATLAB.

Electronics & Hardware: PCB bring-up & rework, STM32, Microchip, TI, I²C, SPI, CAN, FPGA, oscilloscopes, logic analyzers.

Technical Experience

Embedded Software Engineer Intern Jan. 2026 – Sep. 2026
Atomic Semi — Fab2 *San Francisco, CA*

- Brought up STM32 boards and developed bare-metal C/FreeRTOS firmware using DMA, I²C, and CAN, streaming Protobuf telemetry upstream and performing hands-on PCB rework.
- Engineered the full software stack for a wafer storage, cleaning, and delivery tool. Implemented data-driven custom motion profiles to achieve synchronized actuation, passing a **continuous 24-hour** factory acceptance test.
- Implemented a 1 kHz piezo actuator PID control loop using capacitive feedback for **nanometer-scale** mask alignment.
- Owned software for a vacuum-gauge calibration station; used data analysis and molecular-flow research to uncover system limitations and drive hardware and calibration changes improving near-vacuum sensor accuracy.

Firmware & Automation Engineer Co-op Sep. 2024 – Aug. 2025
Analytic Systems *Delta, BC*

- Identified a test bottleneck, pitched an ROI-backed automation plan, and delivered a dual-probe flying-probe system with six I²C-controlled steppers; achieved 0.4 mm-pitch probing and **cut manual testing by 80%**. *Demo*
- Shipped a full-stack PCB test app (GUI, serial DMM/PSU control, DB analytics, PDF reports), adopted for daily use.
- Modernized legacy products by implementing **CAN** firmware mapped to the **NMEA-2000** marine standard.
- Brought up a power telemetry board with ADC sampling, ISR-based alerts, digital averaging, and multiplexed display.

Avionics Lead & Firmware Developer Sep. 2024 – Jan. 2026
UBC Orbit — University Satellite Engineering Design Team *Vancouver, BC*

- Led a 6-member avionics subteam across sensors, actuators, flight software, system integration, and timelines.
- Developed TI MCU firmware with FreeRTOS for flight software; wrote I²C/SPI drivers and tuned ISR priorities.
- Built a **3-axis gimbal** HIL testbed for magnetic noise emulation to validate real-time sensor fusion in MATLAB. *Demo*

Personal Projects

BilliardBot VisionAI *Demo* 2024 – 2025

- Built a **pool-playing robot** with a high-voltage solenoid striker, achieving **90% accuracy** on bank shots over 50+ trials.
- Developed an OpenCV shot-planning pipeline featuring fisheye calibration, ball localization, physics-based ball trajectory simulation, and closed-loop visual servoing.
- Implemented a tunable high-voltage solenoid and control electronics with a 14 V → 200 V boost and opto-isolation.

Miscellaneous: 2022 – 2024

- **Airbus A320 sidestick** *Demo* | Reverse-engineered an actual Airbus A320 sidestick to make it computer-compatible.
- **384 V Multistage Coil Gun** *Demo* | 384 V coil gun with custom IR speedometer, capable of **piercing metal sheets**.
- **Automatic Xylophone** *Demo* | Electromagnetic xylophone with DIY solenoids, 24 transistors, and MIDI control.